DATABASE DESIGN AND DEVELOPMENT (IT 2140)

LECTURE 01- INTRODUCTION TO DBMS, DATABASE DESIGN PROCESS AND DATA-INTENSIVE APPLICATIONS

LECTURE CONTENT

- Define key database-related terminologies
- Introduction to databases, DBMS and their benefits
- Database design process
- Data-intensive applications

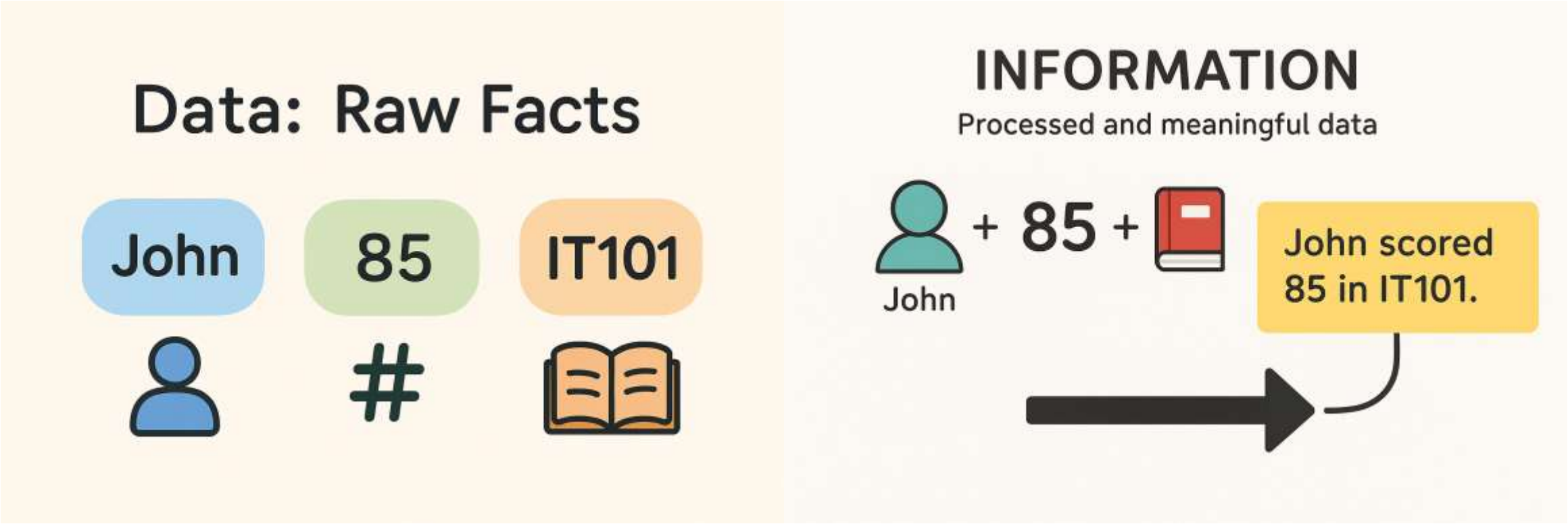
LEARNING OUTCOMES

- Explain what is a database and a DBMS.
- Identify situations where using a database would be beneficial.
- Explain the database design process.
- Identify the importance of DBMSs in data-intensive applications.

WHAT IS DATA VS INFORMATION?

- Data refers to raw, unprocessed facts or values that, on their own, may not carry meaning. It can be in the form of numbers, text, images, or symbols.
- Information is data that has been processed, organized, or structured in a way that makes it meaningful and useful.

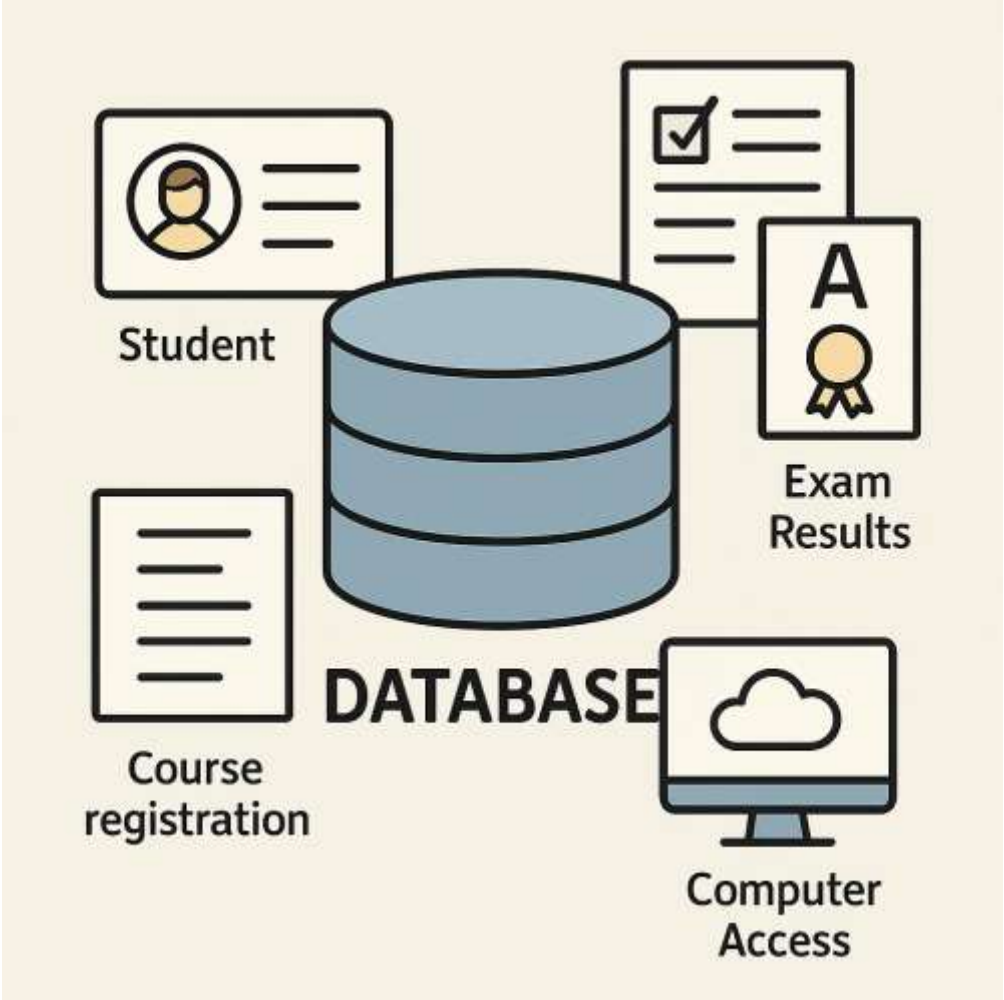
WHAT IS DATA VS INFORMATION?

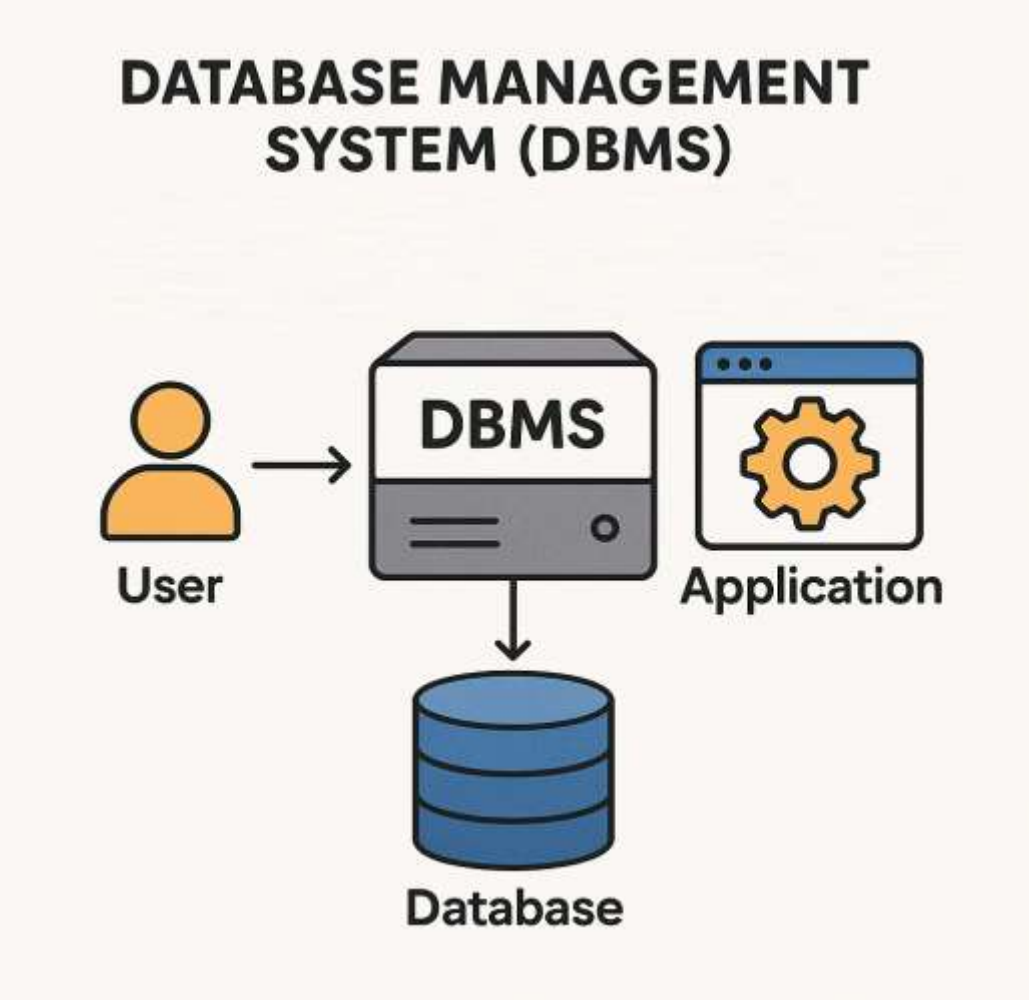


WHAT IS A DATABASE & A DBMS?

- A database is a collection of related data.
- A database management systems (DBMS) is a general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications.
  - What do you understand by defining, constructing, manipulating and sharing databases?

WHAT IS A DATABASE & A DBMS?





? QUESTION FOR STUDENTS

Can you identify one real-world place or system where databases are used? For each example,

Explain:

- What kind of data is stored ?
- Who uses the data ?
- Why a database is necessary in that context?

DATABASE SYSTEM ENVIRONMENT

- Database system environment normally can be considered to have five major parts.
  1. Hardware
  2. Software
  3. People
  4. Procedures
  5. Data

DATABASE APPROACH VS FILE PROCESSING SYSTEM

- Database system is of self-describing nature
- Database enables insulation between programs and data, and data abstraction
- Databases support of multiple views of the data
- Sharing of data and multiuser transaction processing
- Security and access levels

ADVANTAGES OF DATABASE APPROACH

- Data independence – application programs are independent of the way the data is structured and stored
- Efficient data access – make the data retrieval efficient using sophisticated techniques to store and access data
- Enforcing integrity – provide capabilities to define and enforce constraints
  - Ex: Data type for a name should be string
- Restricting unauthorized access – allows only authorized users to access data
- Providing backup and recovery – protects users from the effect of system failures
- Concurrent access – simultaneous access of data by different users

DATABASE DESIGN

Why is Database Design important?

Without careful planning you may create a database that…

- contains unnecessary data which occupies the storage space
- omits data required to create critical reports
- takes a considerable time to respond to user questions
- produces results that are incorrect or inconsistent
- is unable to accommodate changes in the user's requirements

A poorly designed database will require more time in the long-term…!

DATABASE DESIGN PROCESS

- There are six main phases of the process to develop a database
  - Requirement collection and analysis
  - Conceptual database design
  - Logical database design
  - Schema refinement
  - Physical database design
  - Security design

STEP 1: REQUIREMENT COLLECTION AND ANALYSIS

The goals of the requirements analysis are:

- to determine the **data requirements** of the database in terms of objects
- to identify and describe the information about these objects
- to identify the relationships among these objects
- to identify different transactions that will be performed on the database
- to identify performance, integrity, security or administrative constraints to be imposed on the database
- to identify design and implementation constraints if any (ex: specific technologies, hardware and software, programming languages, policies, standards, etc…)

STEP 1: REQUIREMENT COLLECTION AND ANALYSIS

Information needed for the requirements analysis can be gathered in several ways:

- **Review of existing documents –**

Can become familiar with the organization/ activity you need to model by reviewing the existing documentation. Ex: existing forms and reports, written guidelines, job descriptions, personal narratives, etc…

- **Interviews with end users –**

Can organize individual/group meetings with the end users. Can use a blackboard, flip chart, or overhead transparencies to record information gathered from the interviews.

- **Review of existing automated systems –**

Can review the system design specifications and documentation, if the organization already has an automated system

STEP 2: CONCEPTUAL DATABASE DESIGN

- The information gathered in the requirements analysis phase is used to create a:

- high-level description of the data in a conceptual data model or *Semantic Data Model.*

Eg. ER Model

STEP 3: LOGICAL DATABASE DESIGN

- In this step, we determine the DBMS to implement the database & also the data model.
- We utilize the conceptual schema created in the previous step and convert it into a schema of a particular data model* (e.g. Relational Data Model)
- *we will cover this in the next two lectures.

STEP 4: SCHEMA REFINEMENT

- The schema created by the logical database design phase is further refined for potential problems such as redundancies.

STEP 5: PHYSICAL DATABASE DESIGN

- In this step, performance criteria are taken into consideration and further enhancements to the schema & creation of indexes are considered.

STEP 6: SECURITY DESIGN

- Different user groups and their roles are identified.
- Appropriate levels of access are then provided to the data ensuring that users have access to only the necessary data.

Eg. Bank

Customer – read access

Teller – read/update (limited)

Manager – read/update

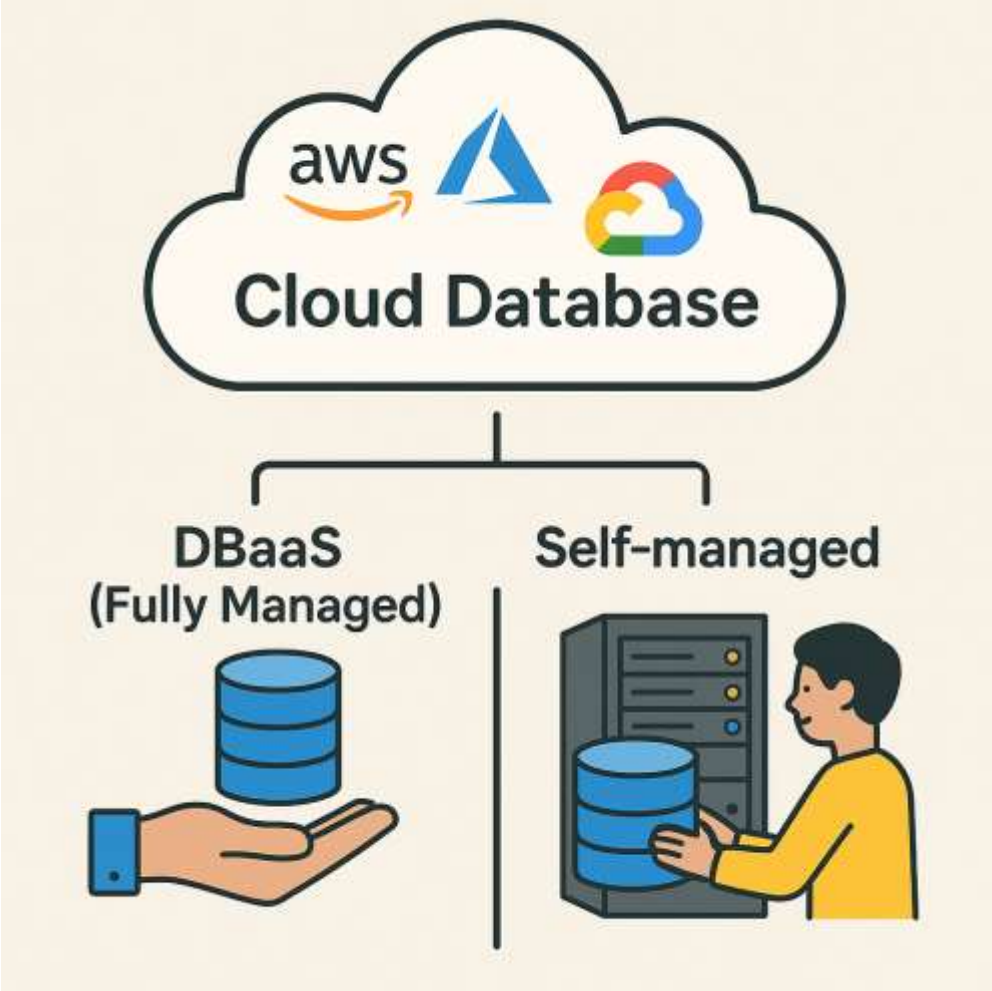
DATABASE HOSTING

- Database hosting defines where and how a database is deployed, stored, and accessed.
- Selecting the appropriate hosting type depends on:
  - Scalability
  - Cost
  - Security
  - Performance

TYPES OF DATABASE HOSTING

- On-Premise Hosting
- Cloud Hosting

WHAT IS CLOUD DATABASE?



EXAMPLES OF CLOUD DATABASE

Cloud Database	Examples
Relational Cloud Databases	AWS RDS, Azure SQL Database, Google Cloud SQL
NoSQL Cloud Databases	MongoDB Atlas, AWS DynamoDB, Google Firestore, Azure Cosmos DB

THINK–PAIR–SHARE ACTIVITY

- Cloud Databases – Advantages and Challenges

DATA-INTENSIVE APPLICATIONS

What are Data-Intensive Applications?

- Focus on handling large volumes of data rather than performing complex computations.
- Involve extensive data storage, retrieval, and processing.
- Require fast and consistent access to data.
- Must be scalable to handle growing data and user loads.
- Common in systems with real-time access and multiple concurrent users.
- Depend heavily on reliable databases and DBMSs for performance and stability.

EXAMPLES OF DATA-INTENSIVE APPLICATIONS

Domain	Example Application	Types of Data Handled
Social Media	Facebook, Instagram	Posts, images, comments, likes, user profiles
E-Commerce	Amazon, eBay	Product listings, transactions, user carts, reviews
Banking	Online banking apps	Customer info, transactions, loan history, statements
Education	Learning Management Systems (LMS), like Moodle	Grades, courses, materials, attendance
Search Engines	Google, Bing	Crawled pages, user queries, ranking metadata

WHY IS A DBMS CRITICAL FOR DATA-INTENSIVE APPLICATIONS?

DBMS Feature	How It Helps
Data Storage & Retrieval	Efficient storage of huge volumes of structured data
Indexing	Fast lookups and search operations
Transactions & Concurrency	Handles simultaneous users and operations safely
Security & Access Control	Ensures authorized access to sensitive data
Backup & Recovery	Prevents data loss in case of failure
Query Optimization	Speeds up processing of complex queries
Scalability (esp. in Cloud)	Supports growth in data, users, and workloads dynamically

HOMEWORK QUESTION

- Identify and describe five (5) modern technologies used in data-intensive applications.
- For each technology, include:
  - The name of the technology
  - Its purpose or function
  - One real-world example where it is used

WHAT YOU HAVE TO DO BY NEXT WEEK

- Try out the self-test questions on the course web.
- Try out tutorial and bring the answers to the class.